

# AI StudyBuddy

AI-Powered Study Management and Learning Support System

## PHASE 1

Brainstorming & Ideation

| Role        | Team Member     |
|-------------|-----------------|
| Team Leader | Kishore T       |
| Team Member | Logeshwaran D   |
| Team Member | Manikandan M    |
| Team Member | Mohamed Safir M |

Submitted as a phase-wise project submission.

# 1. Problem Identification

Students often need to convert large study materials into manageable learning resources. AI StudyBuddy is planned as a web-based study support system that provides a structured backend for student learning and study management.

## 2. Proposed Idea

AI StudyBuddy uses a Node.js and Express.js backend, MongoDB for data storage, JWT-based authentication, and Google Gemini API configuration for AI-assisted learning features.

## 3. Objectives

- Provide a backend platform for student learning support.
- Allow users to register and authenticate securely.
- Store users and study materials using MongoDB.
- Generate learning resources such as summaries, flashcards, quizzes and study plans through AI-enabled APIs.
- Provide administrator capabilities for user management.

## 4. Selected Features

- Student registration and login
- Study material upload
- AI summary generation
- AI flashcard generation
- AI quiz generation
- Personalized study-plan generation
- Administrator registration/login and user management

## 5. Phase Outcome

The project idea and major functional scope are defined. The implementation evidence supplied for later phases is used in the following documents.